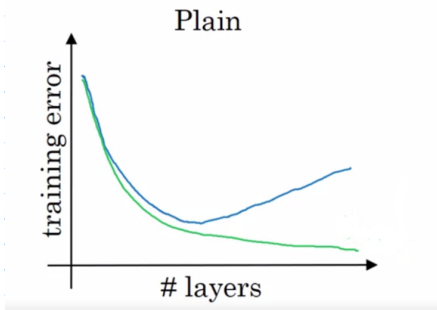


- When building a ConvNet, typically you start with some POOL layers followed by some CONV layers. True/False?
☐ True
☒ False
- In order to be able to build very deep networks, we usually only use pooling layers to downsize the height/width of the activation volumes while convolutions are used with "valid" padding. Otherwise, we would downsize the input of the model too quickly.
☐ True
☒ False
- Based on the lectures, in the following picture, which curve corresponds to the expected behavior in theory, and which one corresponds to the behavior we get in practice? This when using plain neural networks.


The blue one depicts the theory, and the green one the reality.

The green one depicts the results in theory, and the blue one the reality.

The green one depicts the results in theory, and also in practice.

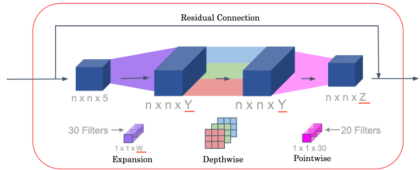
The blue one depicts the results in theory, and also in practice.
- The following equation captures the computation in a ResNet block. What goes into the two blanks above?

$$a^{[l+2]} = g(W^{[l+2]}g(W^{[l+1]}a^{[l]} + b^{[l+1]}) + b^{[l+2]} + \text{_____}) + \text{_____}$$

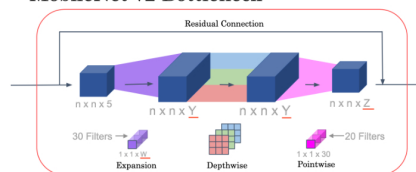
☒ $a^{[l]}$ and 0, respectively

☐ 0 and $a^{[l]}$, respectively

☐ 0 and $z^{[l+1]}$, respectively

☐ $z^{[l]}$ and $a^{[l]}$, respectively
- In the best scenario when adding a ResNet block it will learn to approximate the identity function after a lot of training, helping improve the overall performance of the network. True/False?
☐ True
☒ False
- For a volume of $125 \times 125 \times 64$ which of the following can be used to reduce this to a $125 \times 125 \times 32$ volume?
☒ Use a 1×1 convolutional layer with a stride of 1, and 32 filters.
- Use a 1×1 convolutional layer with a stride of 2, and 32 filters.
- Use a POOL layer of size 2×2 with a stride of 2.
- Use a POOL layer of size 2×2 but with a stride of 1.
- Which of the following are true about bottleneck layers? (Check all that apply)
☐ The bottleneck layer has a more powerful regularization effect than Dropout layers.
- By adding these layers we can reduce the computational cost in the inception modules.
- ☒ Bottleneck layers help to compress the 1×1 , 3×3 , 5×5 convolutional layers in the inception network.
- ☐ The use of bottlenecks doesn't seem to hurt the performance of the network.
- Models trained for one computer vision task can't be used directly in another task. In most cases, we must change the softmax layer, or the last layers of the model and re-train for the new task. True/False?
☒ True
☐ False
- Which of the following are true about Depth wise-separable convolutions? (Choose all that apply)
☒ They combine depthwise convolutions with pointwise convolutions.
- ☒ They have a lower computational cost than normal convolutions.
- ☐ The result has always the same number of channels n_c as the input.
- ☐ They are just a combination of a normal convolution and a bottleneck layer.
- Fill in the missing dimensions shown in the image below (marked W, Y, Z).


MobileNet v2 Bottleneck



- ☐ W = 30, Y = 20, Z = 20
- ☐ W = 5, Y = 20, Z = 5
- ☒ W = 5, Y = 30, Z = 20
- ☐ W = 30, Y = 30, Z = 5

